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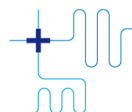
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The use of pressure controlled Fabry-Pérot interferometer with linear scanning of data for Brillouin-type experiments

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The article presents results from work with Fabry-Pérot interferometers in Brillouin laser light scattering experiments, where optical signals of very low level intensity are observed. The information presented here can be useful in other types of optical experiments where scanning in the Fabry-Pérot interferometer spectral range has to be used. In such situations the shape of spectral lines as well as their relative distances can be detected. The key to the solution presented here is the use of a silicon-membrane pressure sensor coupled to a pressure chamber. It makes it possible to view spectral lines equally spaced after nonlinear flow of air from a chamber where the Fabry-Pérot interferometer is placed. Linear scanning in the spectral range equal to a frequency of about 150 GHz is possible. The method can be applied to Fabry-Pérot's etalons, very frequently produced some years ago. Now it should find new fields of application, in a simple and cost effective way, in student laboratories as well as in other research institutions. © 2000 American Institute of Physics. [S0034-6748(00)01808-6]

I. INTRODUCTION

The Fabry-Pérot interferometer is widely applied in different areas of spectroscopic research, especially in Brillouin spectroscopy where high contrast and resolving power have to be used. Studies using this kind of spectroscopy developed rapidly in the early 1960s when lasers became commercially available.

The Fabry-Pérot interferometer in Brillouin spectroscopy is used in configurations where there exists a need to evaluate the shape of spectral lines as well as their relative distances in the frequency domain. It is achieved by optical path changes for optical rays trapped in an interferometer cavity. In this way an interference pattern moves in respect to a photomultiplier gap placed in the fixed point of a space. In such situations we deal with a so-called scanning of the Fabry-Pérot interferometer. Two main methods of scanning exist. The first one utilizes index of refraction changes of the gas (air) surrounding the interferometer cavity, and the second one applies space-distance changes between interferometer mirrors.

In the last 40 years some interesting scanning setups have been constructed. The oldest from the early 1960s¹ is a pressure-controlled solution. By the use of a very sophisticated set of chambers and valves linear scanning was achieved. However, this very complexity was a disadvantage. Most other solutions have been based on piezoelectrically controlled distance between mirrors, as in the tandem setup of Mock, Hillebrands, and Sandercock.² In the Pakulski and Holuj³ spectrometer spectra from two samples, a measured one and a reference have been recorded simultaneously by the use of optical fibers. In another, the Vachner, Sussner, and Schickfus,⁴ construction, flat and confocal mir-

rors are combined. The commercially available multi-pass setup from Burleigh Inc. is well known. Finally, the most advanced is the fully automated and self-aligning multi-pass tandem Fabry-Pérot interferometer proposed recently by Hillebrands.⁵

In the current article a return into an old method is proposed — pressure controlled interferometry with linear scanning, as opposed to the previous design, is obtained in a very simple way. It is achieved, by the use of silicon-membrane pressure sensors, which provide voltage signals linearly proportional to the pressure in the chamber where the interferometer is placed. The proposed solution can be applied to old models of Fabry-Pérot interferometers (etalons), where piezoelectric scanning is not always possible.

II. THEORETICAL BACKGROUND TO THE IDEA

It is not difficult to prove that the refractive index of an ideal gas, to a very good approximation, is a linear function of a pressure. It will be shown below what the consequences of the above dependence are for an interference pattern produced by the Fabry-Pérot interferometer.

For normal incidence of light flux at interferometer mirrors the relation between the mirror space d , the refractive index of air n , the number of the order in an interference pattern m , and the light wavelength passing through an interferometer λ is as follows:^{6,7}

$$2 \cdot d \cdot n = m \cdot \lambda. \quad (1)$$

Pressure changes cause refractive index changes. We can express this fact mathematically by the differential operation in Eq. (1), namely

$$2 \cdot d \cdot dn = dm \cdot \lambda. \quad (2)$$

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This trivial calculus describes the situation where interference fringes, created by an interferometer, move as a result of pressure changes.

Then, we can start from the well-known Clausius–Mossotti equation⁸

$$\frac{\varepsilon - 1}{\varepsilon + 2} = \frac{n_k \alpha}{3\varepsilon_0}, \quad (3)$$

where ε is the relative permittivity of a gas, n_k is the gas concentration, α is the average polarizability of a gas, and ε_0 is the permittivity of free space (vacuum). Next by some simple approximations, namely

$$\varepsilon \cong 1, \quad \varepsilon + 2 \cong 3, \quad n \cong 1, \quad n + 1 \cong 2, \quad (4)$$

we obtain the relation between interference-pattern-order changes caused by pressure changes in the following form:

$$dm = \frac{2d}{\lambda} \cdot \frac{n_0 - 1}{p_0} dp. \quad (5)$$

The above equation provides information about the number of the orders changed along with pressure changes. This fact is expressed once more by integrating Eq. (5), namely

$$\int_{m_1}^{m_1+m} dm = \int_0^p \frac{2d}{\lambda} \cdot \frac{n_0 - 1}{p_0} dp. \quad (6)$$

After the above integration we obtain a linear dependence between interference orders, scanned from the order m_1 to the order $m_1 + m$, and a pressure

$$m = \frac{2d}{\lambda} \cdot \frac{n_0 - 1}{p_0} p. \quad (7)$$

The transmission function⁶ of the Fabry–Pérot interferometer, with omitted factor responsible for losses, can be now expressed as a function of pressure

$$t(p) = \frac{1}{1 + \frac{4R}{(1-R)^2} \cdot \sin^2\left(\frac{2d}{\lambda} \cdot \frac{n_0 - 1}{p_0} p \pi + \varphi\right)}, \quad (8)$$

where R is the reflection coefficient of the interferometer mirrors, and φ possesses the sense of the resultant difference of the phase between recombining wave fronts of an optical signal, passing through the interferometer, in a vacuum, when pressure is equal to zero. The transmission function expresses the situation, in which we obtain a periodic interference pattern moving along with pressure changes.

At the end of this chapter it will be convenient to show the exact dependence between the refractive index of a gas and its pressure

$$n = \left(\frac{1 + p \frac{n_0^2 - 1}{p_0(n_0^2 + 1)}}{1 - p \frac{n_0^2 - 1}{p_0(n_0^2 + 1)}} \right)^{1/2}. \quad (9)$$

It must be said that the error caused by the above described approximations does not influence experimental results—its value is on the $9 \times 10^{-7}\%$ level.

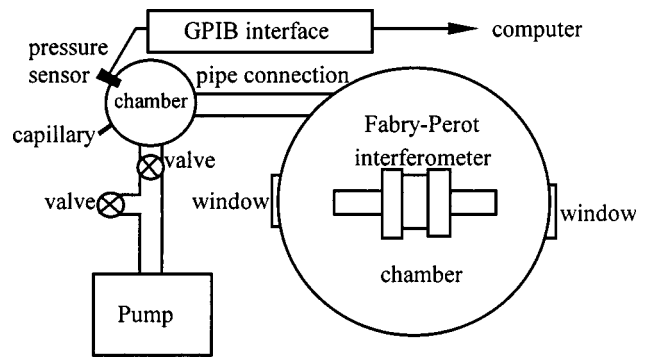


FIG. 1. Schematic of the pressure scanned Fabry–Pérot system.

III. TRANSMISSION FUNCTION. CONSTRUCTION DETAILS AND EXPERIMENTAL ERRORS

Tested in the laboratory, the pressure scanning system consisted of two pressure chambers. An interferometer was placed in the first one. In the second chamber the valves, pressure sensor, and capillary were mounted. The setup is schematically shown in Fig. 1. A separate chamber for control elements is very useful, because wires do not interfere with opening and closing of an etalon chamber. The cham-

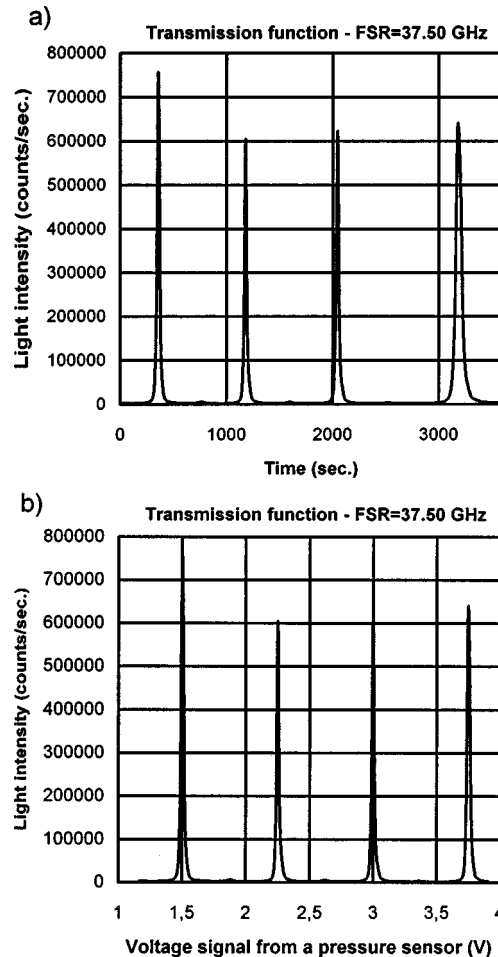


FIG. 2. The transmission function of the Fabry–Pérot interferometer as a function of time (a) where peaks are not equally spaced, and as a function of the voltage signal from a silicon-membrane sensor (b). The full spectral range of the interferometer is equal to 37.50 GHz. Four interference orders were detected.

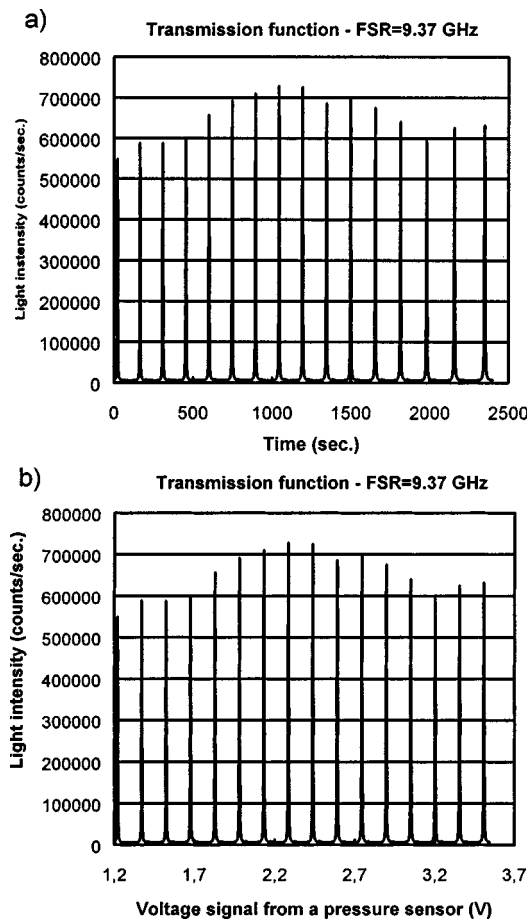


FIG. 3. The transmission function of the Fabry-Pérot interferometer as a function of time (a), and as a function of the voltage signal from a silicon-membrane sensor (b). The full spectral range of the interferometer is equal to 9.37 GHz. Sixteen interference orders are visible. A nonlinear gas flow is strongly visible only for last interference orders.

bers were connected by a rubber pipe of 2 cm diameter and 1.5 m length. Such connections do not influence any impedance effects during gas flow. The diameter of the chambers was 0.25, and 0.60 m, for the control and etalon chamber, respectively. The height of each chamber was equal to 0.25 m. The equipment used in measurements also included a single-mode argon ion laser working at 514.5 nm wavelength, with the power equal to about 100 mW, using the single-pass option Fabry-Pérot interferometer. The unit produced by Photon Inc. for the low-level intensity light detection by the photon counting method⁹ was also used. The signals from the photomultiplier along with pressure were detected at 1 s intervals. Typical duration of measurement was 1 h. A time resolution equal to 1 s generates a frequency error equal to about 0.04 GHz, if three full spectral ranges, each equal to 37.50 GHz, were scanned during 1 h. Etalons for experiments were manufactured many years ago from invar or quartz with 10^{-6} m accuracy. For example, typical values of etalon widths are as follows (in millimeters): 1.004, 3.993, 16.003, or 8.000. The full spectral range (in GHz) can be estimated from a well-known formula as

$$\text{FSR} = \frac{300}{2d}, \quad (10)$$

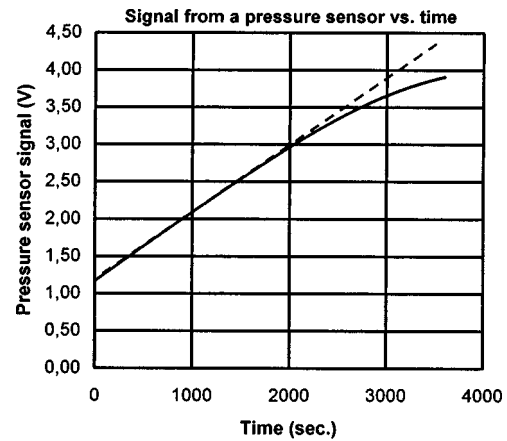


FIG. 4. Signals from a pressure sensor as a function of time. Dashed straight line was added for convenience. An amount of nonlinearity depends on capillary diameter applied.

where the etalon width d is given in millimeters. The value 300 derives from the speed of light, of course. If we do not substitute the exact value for light speed, then we produce an additional error in full spectral range (FSR) equal to about 0.01 GHz.

One of the most important quantities describing the quality of a pressure sensor is the pressure hysteresis error. In our case the Siemens KPY 43 MA sensor was applied. The sen-

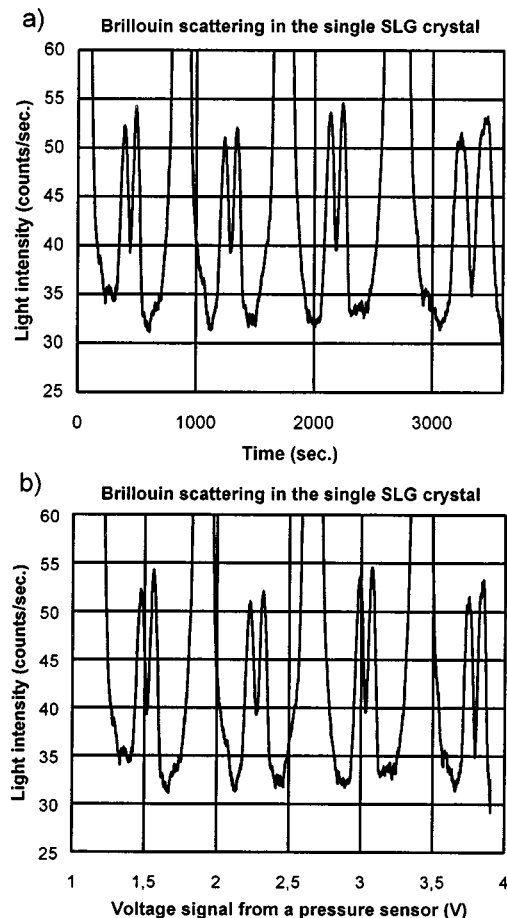


FIG. 5. Brillouin scattering in the single SrLaGaO_4 crystal as a function of time (a) and pressure (b). The crystal possesses tetragonal symmetry. Small optical signals resulting from quasitransverse acoustic waves are seen.

sor detects the absolute value of the atmosphere pressure. The pressure hysteresis error was equal to 0.1%,¹⁰ however it was not important during measurements, where pressure changes have gone into one direction only.

The above sensor, as well as others available commercially, possesses active thermal stabilization. The error caused by this effect in our case was not larger than 0.095% K⁻¹.

Another source of experimental error is the nonlinearity between output voltage from the sensor and the pressure transformed to this voltage signal. The nonlinear error is of the order of 0.15% for the sensor applied.

The next source of error is the accuracy of the Fabry-Pérot etalon width. In our case it caused an error of 0.01 GHz for the FSR equal to 37.50 GHz. The maximum error of this type was equal to 2 GHz, for the FSR=500 GHz, and minimum value was equal to 0.0002 GHz for the FSR=5.0005 GHz. Of course all errors depend on the quality of manufactured etalons.

Few pictures show transmission function as the function of time and pressure for the FSR=37.50 GHz [Figs. 2(a) and 2(b)] and FSR=9.37 GHz [Figs. 3(a) and 3(b)]. The dependence between pressure and time is shown in Fig. 4. It is not difficult to recognize some regions of almost linear dependence. On the other hand, the character of the presented plots depends on the diameter of the capillary through which a gas flows into chambers.

At the end of this chapter some Brillouin spectra from the single SrLaGaO₄ (SLG) crystal are presented in the same manner—as a function of time, and as a function of pressure—in Figs. 5(a) and 5(b), respectively. It is seen that signals presented in the voltage domain possess equal widths

for different interference orders. The measured value of the FSR derived from this spectrum is equal to 37.50±0.04 GHz. The registered frequency of the Brillouin peak resulted from a quasitransverse wave is equal to 20.9±0.1. The provided values are valid for the level of confidence (standard deviation) equal to 0.7.

It is clear that this approach to linearization leads to a nonuniform scan rate, however the rate is very slow during the scanning process. In our example observed Brillouin peaks have been evaluated from 185 data points in the first FSR, and from 205 points in the second FSR. This gives a very good accuracy, which can be additionally improved by increasing the detection interval ten times or larger depending on what computer equipment is applied.

ACKNOWLEDGMENT

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